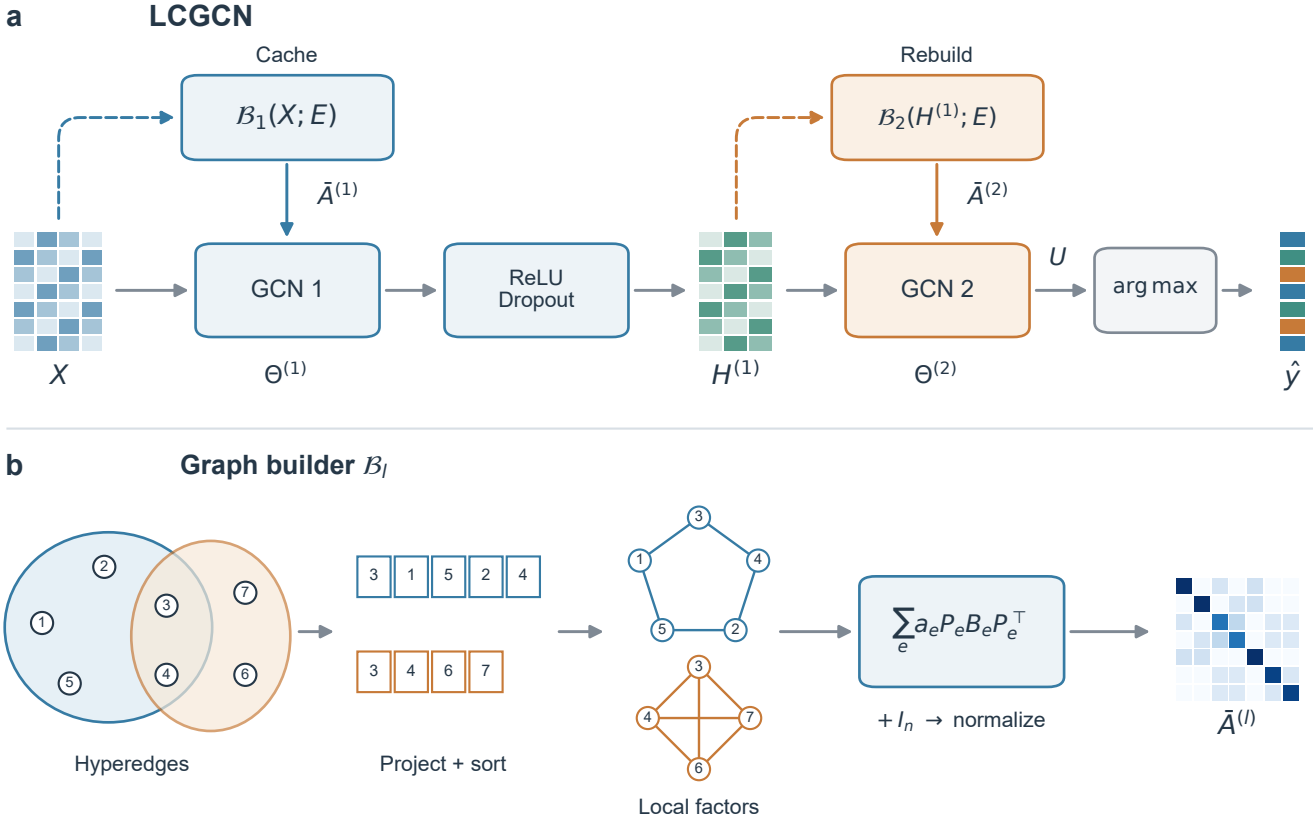

a
LCGCN
Cache
$\mathcal{B}_1(X;E)$
$\bar{A}^{(1)}$
GCN 1
ReLU
Dropout
Rebuild
$\mathcal{B}_2(H^{(1)};E)$
$\bar{A}^{(2)}$
GCN 2
U
arg max
X
$\Theta^{(1)}$
$H^{(1)}$
$\Theta^{(2)}$
$\hat{y}$
b
Graph builder $\mathcal{B}_l$
Hyperedges
Project + sort
Local factors
$\sum_e a_e P_e B_e P_e^\top$
$+ I_n \rightarrow$ normalize
$\bar{A}^{(l)}$